RadixMaths

A Level Mathematics

Practice Paper 1

Pure Mathematics

Level: **Standard**

Similar in difficulty to one of the more accessible real A Level papers.

Time: 2 hours **Total marks: 100**

For students of every exam board. This paper covers only the pure mathematics that the Department for Education requires in every A Level Mathematics course, so it suits AQA (7357), Edexcel (9MA0), OCR A (H240) and OCR B MEI (H640). Questions may use content from Year 12 or Year 13.

You will need: a calculator, your exam board's formulae booklet, and paper to write on.

Instructions

- Answer **all** the questions, and label your answers to each part clearly.
- Show enough working to make your method clear. Answers without working may not gain full credit.
- Where a question says *show detailed reasoning*, or says that solutions relying on calculator technology are not acceptable, you must show every step of your working.
- Give inexact answers to three significant figures unless the question says otherwise. Use radians unless a question uses degrees.
- Diagrams are not drawn to scale.

Information

- The total mark for this paper is 100. The marks for each part are shown in brackets, for example **(3)**. Use them as a guide to how long to spend.
- The mark scheme and full worked solutions are in a separate document. Try the whole paper under timed conditions before you look at them.

1. Given that θ is small and measured in radians, which of these is an approximation for $\cos 3\theta$?

Tick **one** box.

☐ $1 - \frac{3\theta^2}{2}$ ☐ $1 - \frac{9\theta^2}{2}$ ☐ $1 - 9\theta^2$ ☐ $1 - 3\theta$

(1)

(Total for Question 1 is 1 mark)

2. Find

$$\int \left(6x^2 - \frac{4}{x^3} + 3\sqrt{x}\right) \mathrm{d}x$$

giving each term in its simplest form. **(4)**

(Total for Question 2 is 4 marks)

3.

$$\mathrm{f}(x) = 2x^3 - 3x^2 - 11x + 6$$

(a) Use the factor theorem to show that $(x - 3)$ is a factor of $\mathrm{f}(x)$. **(2)**

(b) Hence factorise $\mathrm{f}(x)$ completely. **(2)**

(c) Hence solve the equation

$$2\mathrm{e}^{3t} - 3\mathrm{e}^{2t} - 11\mathrm{e}^{t} + 6 = 0$$

giving your answers as exact logarithms. **(2)**

(Total for Question 3 is 6 marks)

4. (a) Find the first three terms, in ascending powers of x, of the binomial expansion of

$$(2 - x)^6$$

giving each term in its simplest form. **(3)**

(b) Hence find the coefficient of x^2 in the expansion of

$$(3 + 2x)(2 - x)^6$$

(2)

(Total for Question 4 is 5 marks)

5. (a) Solve the equation

$$\log_2(x + 1) - \log_2 x = 3$$

(3)

(b) Solve the equation

$$5^{2x-1} = 7^x$$

giving your answer to 3 significant figures. **(3)**

(Total for Question 5 is 6 marks)

© 2026 RadixMaths • www.radixmaths.co.uk

6. The circle C has equation

$$x^2 + y^2 - 6x + 4y - 12 = 0$$

(a) Find

(i) the coordinates of the centre of C,

(ii) the radius of C. **(3)**

(b) The point $P(6, 2)$ lies on C.

Find an equation of the tangent to C at P, giving your answer in the form $ax + by + c = 0$, where a, b and c are integers. **(3)**

(Total for Question 6 is 6 marks)

7. Figure 1 shows a sector OAB of a circle with centre O and radius r cm. The angle AOB is 1.2 radians. The perimeter of the sector is 32 cm.

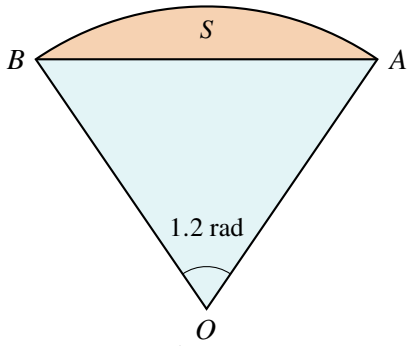


Figure 1

(a) Show that $r = 10$. **(2)**

(b) Find the area of the sector OAB. **(1)**

The segment S, shaded in Figure 1, is bounded by the chord AB and the arc AB.

(c) Find the area of S, giving your answer to 3 significant figures. **(3)**

(Total for Question 7 is 6 marks)

8. The curve C has equation

$$y = x^3 - 6x^2 + 9x + 2$$

(a) Use calculus to find the coordinates of the stationary points of C, and determine the nature of each one. **(5)**

(b) Hence find the range of values of the constant k for which the equation

$$x^3 - 6x^2 + 9x + 2 = k$$

has three distinct real solutions. **(2)**

(Total for Question 8 is 7 marks)

© 2026 RadixMaths • www.radixmaths.co.uk

9. Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ when

(a) $y = (3x - 1)^5$ **(2)**

(b) $y = x^2 \ln x, \quad x > 0$ **(2)**

(c) $y = \dfrac{\sin 2x}{x}, \quad x \neq 0$ **(2)**

(Total for Question 9 is 6 marks)

10. The functions f and g are defined by

$$\mathrm{f}(x) = 3 + \frac{2}{x - 1}, \qquad x \in \mathbb{R},\ x > 1$$
$$\mathrm{g}(x) = x^2 + 1, \qquad x \in \mathbb{R}$$

(a) State the range of f. **(1)**

(b) Find $\mathrm{f}^{-1}(x)$, stating its domain. **(3)**

(c) Find the value of $\mathrm{gf}(3)$. **(2)**

(Total for Question 10 is 6 marks)

11. *In this question you must show detailed reasoning. Solutions relying entirely on calculator technology are not acceptable.*

(a) Solve, for $0° \leqslant x < 360°$, the equation

$$2\cos^2 x + 3\sin x = 3$$

(4)

(b) Solve, for $0° \leqslant x < 180°$, the equation

$$\tan(2x - 30°) = \sqrt{3}$$

(3)

(Total for Question 11 is 7 marks)

12. Priya is training for a long-distance run. She runs 2 km on day 1 of her training. Each day after that she runs 0.5 km further than the day before.

(a) Find the distance she runs on day 20. **(1)**

(b) Find the total distance she runs in the first 20 days. **(2)**

(c) Find the first day on which the total distance she has run since day 1 is more than 500 km. **(3)**

(Total for Question 12 is 6 marks)

© 2026 RadixMaths • www.radixmaths.co.uk

13. (a) Prove that the sum of the squares of any two consecutive odd numbers is 2 more than a multiple of 8. **(4)**

(b) Show, by means of a counter-example, that the following statement is false.
"$2^n + 1$ is prime for every positive integer n." **(1)**

(Total for Question 13 is 5 marks)

14. (a) Find

$$\int x\,e^{2x}\,dx$$

(3)

The curve with equation $y = 4x - x^2$ and the line with equation $y = x$ intersect at the origin O and at the point A.

(b) Use algebraic integration to find the exact area of the finite region bounded by the curve and the line. **(4)**

(Total for Question 14 is 7 marks)

15. The curve C has parametric equations

$$x = 3\cos t, \qquad y = 2\sin t, \qquad 0 \leqslant t < 2\pi$$

(a) Find a Cartesian equation for C. **(2)**

(b) Find the exact value of $\dfrac{dy}{dx}$ at the point P where $t = \dfrac{\pi}{3}$. **(3)**

(c) The tangent to C at P meets the x-axis at the point Q. Find the x coordinate of Q. **(2)**

(Total for Question 15 is 7 marks)

16.

$$f(x) = x^3 - 3x - 5$$

The equation $f(x) = 0$ has a single real root α.

(a) Show that α lies between 2 and 3. **(2)**

(b) Using the iteration formula

$$x_{n+1} = \sqrt[3]{3x_n + 5}, \qquad x_1 = 2$$

find the values of x_2 and x_3, giving your answers to 4 decimal places. **(2)**

(c) Taking $x_1 = 2.3$ as a first approximation, apply the Newton–Raphson method once to $f(x)$ to find a second approximation for α. Give your answer to 4 decimal places. **(3)**

(Total for Question 16 is 7 marks)

© 2026 RadixMaths • www.radixmaths.co.uk

17. A cup of tea is made with water at 90°C and left in a room where the temperature is a constant 20°C. The temperature of the tea, θ °C, t minutes after it is made is modelled by the differential equation

$$\frac{d\theta}{dt} = -k(\theta - 20)$$

where k is a positive constant.

(a) Solve the differential equation to show that

$$\theta = 20 + 70e^{-kt}$$

(4)

(b) After 5 minutes the temperature of the tea is 55°C. Find the exact value of k. **(2)**

(c) Find how long it takes for the tea to cool to 30°C, giving your answer to the nearest tenth of a minute. **(2)**

(Total for Question 17 is 8 marks)

TOTAL FOR PAPER IS 100 MARKS

END OF QUESTIONS

© 2026 RadixMaths • www.radixmaths.co.uk